TAUREX DRILL BITS
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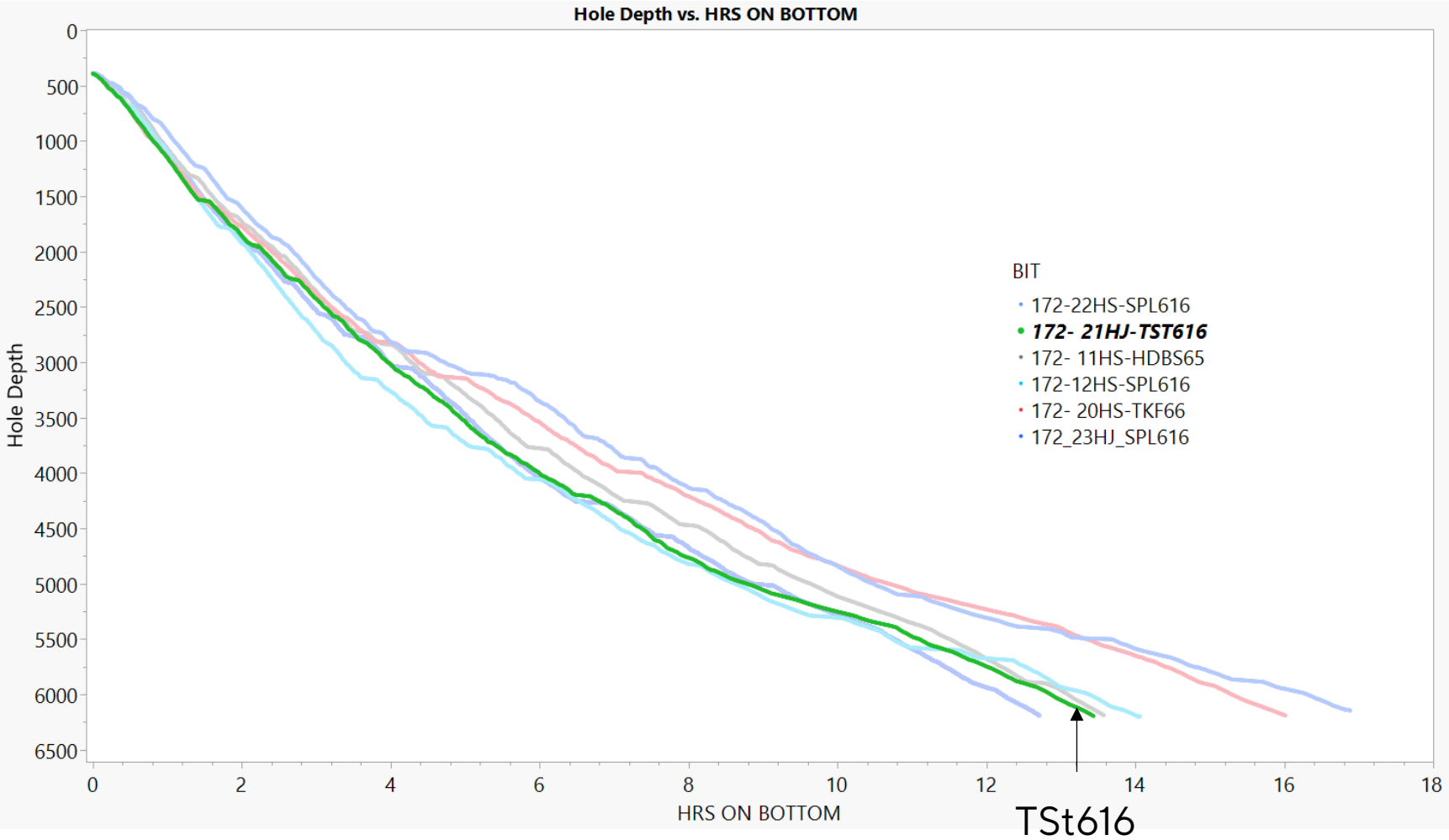
10.625″ Cactus 172 Analysis

Midnight Bourbon

Cactus 171 and Cactus 172 Overview

Hole Depth Vs On-Bottom Hrs

10.625" Midnight Bourbon: Cactus 172



Pad Comparison: On-Bottom Hrs

- On Cactus 172's Midnight Bourbon pad, Taurex TSt616 completed the 10.625" interval in 13.4 hrs
- Taurex delivered the fastest interval on the pad when comparing manufacturer averages:

MFG	#Hrs (Average)
Taurex	13.4
Ulterra	14.4
NOV	15.9
HDBS	13.5

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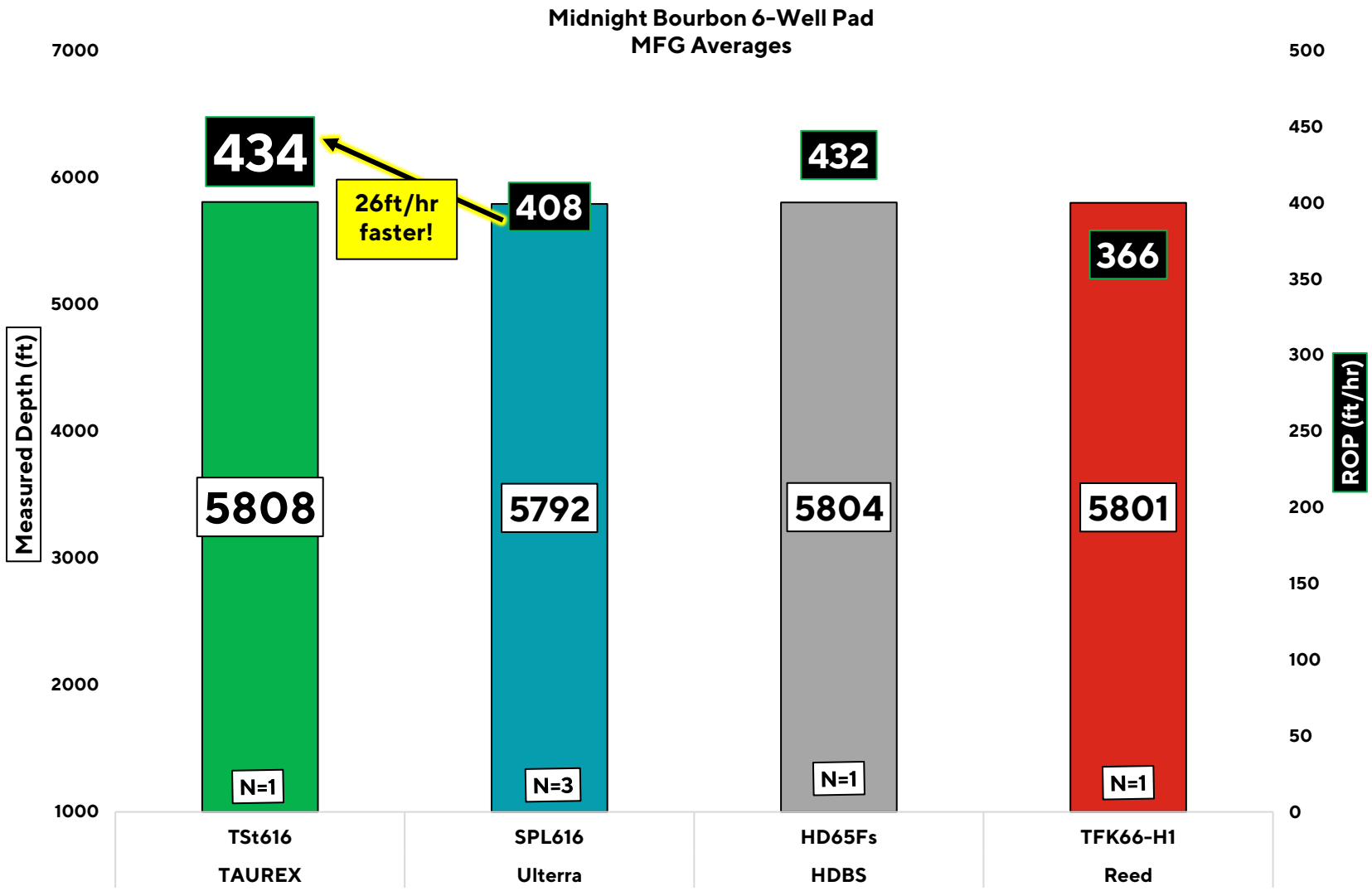
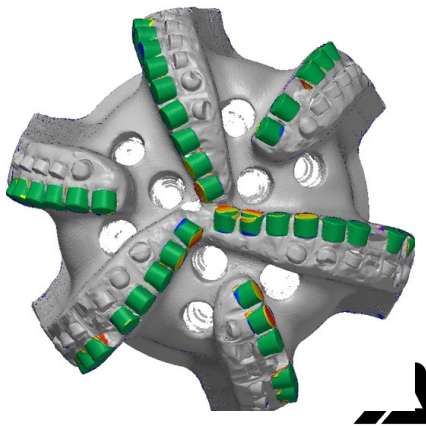
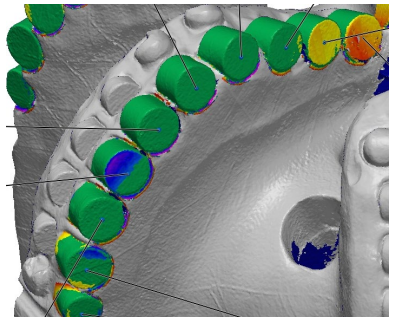
Cactus 172: 10.625" Midnight Bourbon Pad Comparison



Midland County, TX

- On a 6-Well Pad with Cactus 172, Taurex TSt616 drilled the **5,808 ft in 13.4 hrs, averaging 434 ft/hr!**
- TSt616 beat the pace-setting mfg average **by 26ft/hr**
- BitVision Dull Grade: **0.5, 1.2**

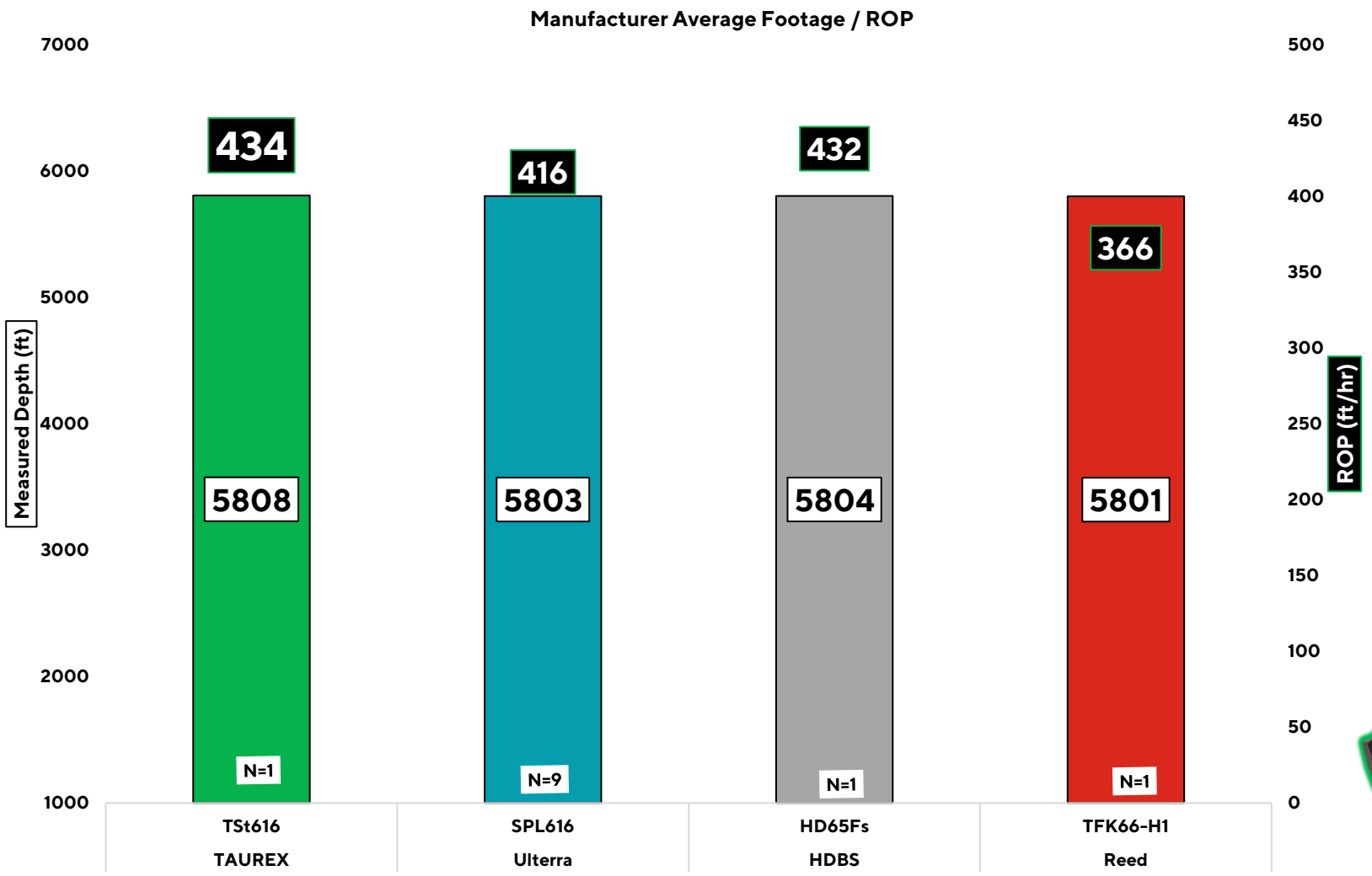




10.625″ TSt616 Run Recap

OPERATOR	RIG	RUN DATE	DEPTH IN	DEPTH OUT	DISTANCE DRILLED
APACHE CORPORATION	Cactus 172		390	6197	5807
WELL	**COUNTY**	**SERIAL**	**HOURS**	**ROP**	**DULL**
Midnight Bourbon 21HJ	Midland , TX	S15728	13.39	433.7	0.5-1.2-BT-S-X-0-LN-TD

Cactus 172 and Cactus 171: Midnight Bourbon Pad Comparison



Midland County, TX

- Across two rigs on the same lease, Taurex TSt616 delivered the fastest 10.625" interval average by manufacturer
- BitVision Dull Grade: **0.5, 1.2**

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Midnight Bourbon Cactus 171 & 172: Runs By Well

★ EDR parameter analysis included

Data Analysis

- The Midnight Bourbon well comparison from Cactus 171/172 uses data from Corva Metrics (10.625" Intermediate Hole Section)
- The runs marked by a ★ are used for additional EDR parameter analysis:
  - Cactus 171 -10HS (SPL616)
  - Cactus 172 – 21HJ (TST616)
- Rotating/Sliding ROPs and percentages for the two wells are provided below: (from Corva)

Rig - Well	Bit	Rotating ROP	Sliding ROP	Rotating Depth (%)	Rotating Time (%)
Cactus 171 - 10HS	SPL616	627	81	95%	73%
Cactus 172 - 21HJ	TSt616	448	140	98%	95%

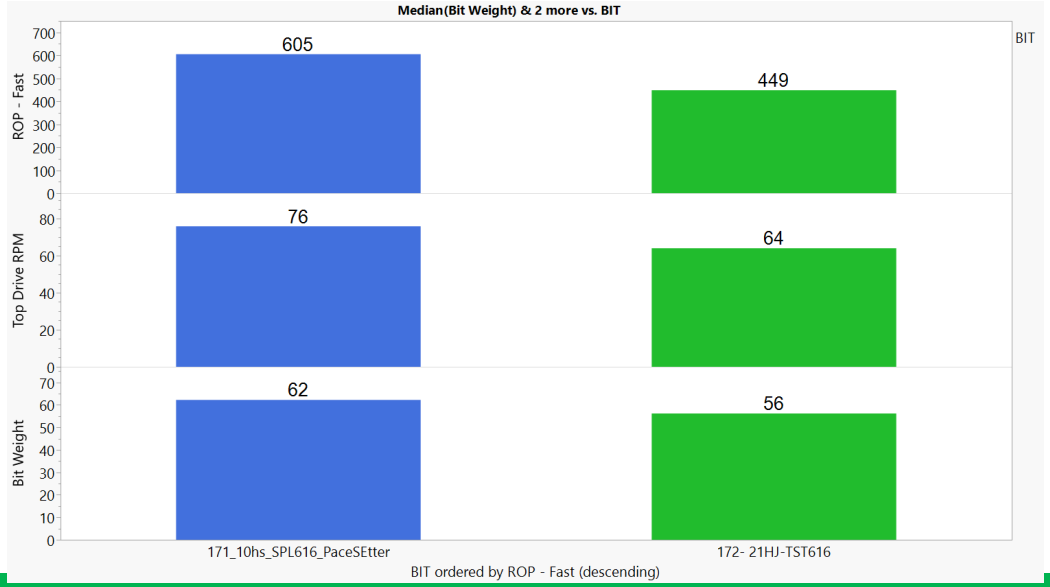
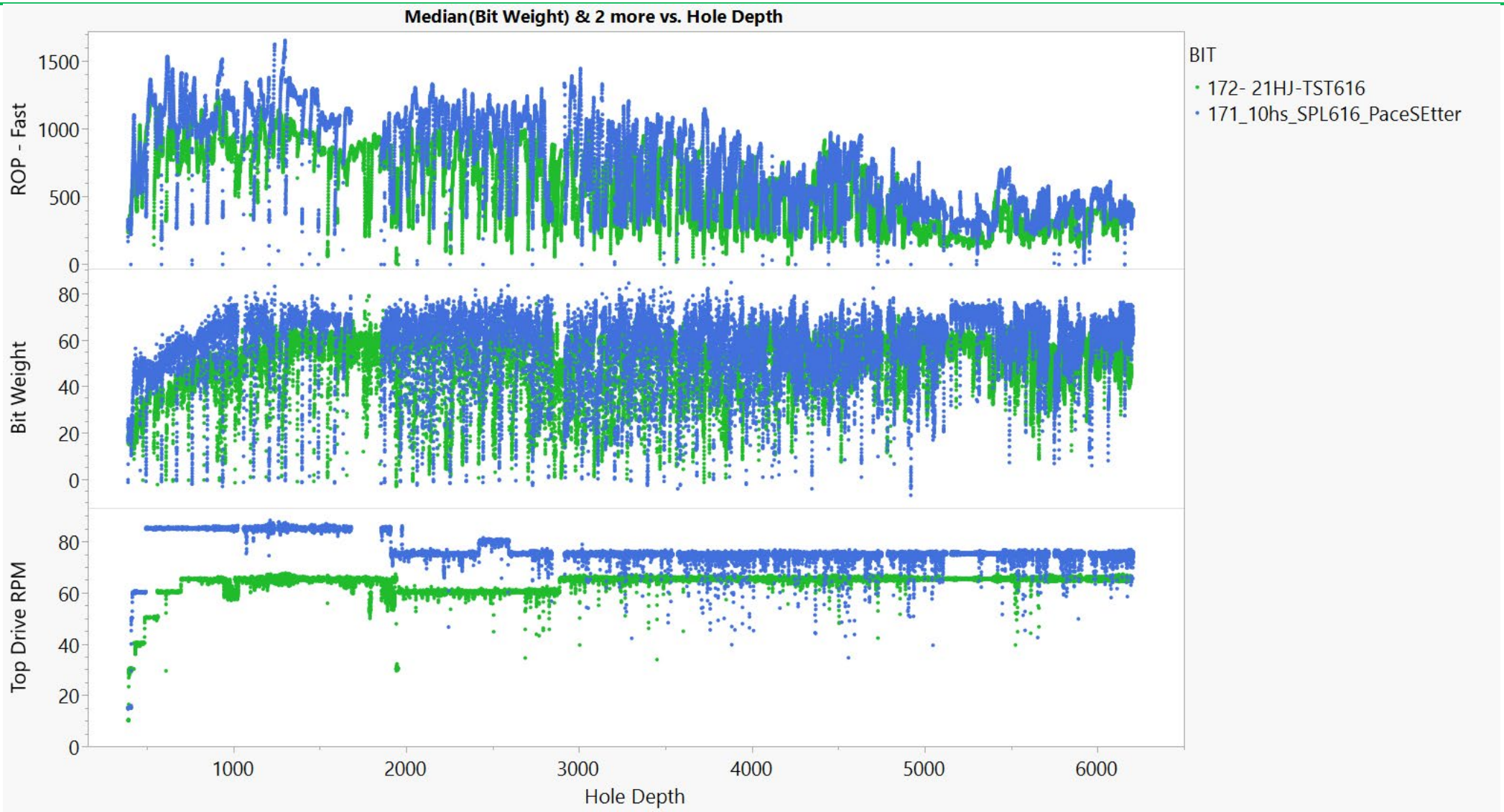
- EDR analysis shows higher rotating WOB and rotary RPM on the 10HS well compared with the 21HJ (TST616) well (next slide)

EDR Analysis: Parameters by Target Formation:

		171-10HS- SPL616	172- 21HJ-TST616
Santa Rosa (886' - 1748')	**Mean ROP**	**1025**	**818**
	Mean Top Drive RPM	**82**	**61**
	Mean WOB (klbs)	**54**	**45**
Rustler/Yates (1748'-2980')	**Mean ROP**	**879**	**630**
	Mean Top Drive RPM	**76**	**61**
	Mean WOB (klbs)	**57**	**46**
Seven Rivers (2980'-4281')	**Mean ROP**	**603**	**467**
	Mean Top Drive RPM	**75**	**65**
	Mean WOB (klbs)	**55**	**49**
Queen (4281'-4415')	**Mean ROP**	**486**	**436**
	Mean Top Drive RPM	**75**	**65**
	Mean WOB (klbs)	**54**	**53**
Grayburg (4415' - 4825')	**Mean ROP**	**534**	**424**
	Mean Top Drive RPM	**74**	**65**
	Mean WOB (klbs)	**56**	**55**
San Andres (4825'- 6035')	**Mean ROP**	**385**	**257**
	Mean Top Drive RPM	**75**	**65**
	Mean WOB (klbs)	**62**	**55**
Glorieta (6031'- 7085')	**Mean ROP**	**404**	**332**
	Mean Top Drive RPM	**75**	**65**
	Mean WOB (klbs)	**62**	**54**

EDR Data Analysis

- The chart on the left provides rotating, on-bottom data separated by target formation and color coded from highest to lowest (red to green) values
  - In all formations, 10HS had the highest WOB, RPM and ROP
- The graph below provides the interval averages:
  - 19% less top drive RPM, and 10% less WOB on the TSt616 (21HJ) well compared with the 10HS well



EDR Overview Of 22HS, 21HJ, and 10HS Wells on the Midnight Bourbon Pad